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Departamento de Engenharia Informática

Segurança em Tecnologias da Informação, **MEI 2022/2023**

Exame – Época Normal (1h45)

14 Jun 2023

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1. Consider email security solutions used on the Internet. What distinguishes PGP and S/MIME in the way they implement authentication and non-repudiation for transmitted messages? Justify your answer.

2. Consider the Feistal structure used in symmetric block encryption algorithms. What are the characteristics of this structure that guarantee the principle of diffusion? Justify your answer.

3. Consider the usage of the CTR (counter) mode for symmetric encryption. Do you consider this mode suitable for encrypting a communication session with a high throughput? Justify your answer.

4. Assume that Alice and Bob intend to communicate securely using asymmetric encryption with the RSA algorithm. Also consider that Alice chooses the values “p=3” and “q=7” as prime numbers, and Bob the values “p=2” and “q=5”. Calculate the public and private keys for Alice and Bob. Then encrypt a message represented by the value “m=9” using Bob's public key.

5. Consider the IPSec protocol, in particular the AH (Authentication Header) header. Is it feasible to use this security header to secure transport mode communications and simultaneously employ SNAT (Source NAT) on a router between the source and destination hosts? Justify your answer.

6. Compare certificate revocation lists (CRL) and the OCSP protocol as methods to verify the revocation status of certificates in the X.509 public key infrastructure (PKI). Discuss the advantages and disadvantages of each method in respect to scalability, efficiency in updating the revocation state information, and ease of use. Justify your answer.

7. Consider the RSA Protocol. Describe how this protocol introduces the concept of the “one-way trapdoor function” and its role in achieving security. Justify your answer.